

RAID Levels

1. Choice of RAID Level

- **Monetary Cost:** Balance between performance and storage overhead.
- **Performance (Normal Operation):** Number of I/O operations per second (IOPS) and bandwidth.
- **Performance During Failure:** How the array behaves if a disk fails.
- **Rebuild Performance:** Time and efficiency in reconstructing failed disks.

1.1 RAID Level Comparisons

RAID 0 – Striping

- High speed but **no safety**.
- Used when data can be quickly regenerated (e.g., scratch disks).

RAID 1 – Mirroring

- Requires **2 block writes**, but faster writes than RAID 5.
- High storage cost (50% utilization).
- Best for workloads with many **random/small updates**.

RAID 5 – Striping + Parity

- Each write needs 2 block reads + 2 block writes.
- Efficient storage utilization: $(n - 1)/n$.
- Best for **large sequential writes**.

RAID 6 – Dual Parity

- Tolerates **two disk failures**.
- Stronger protection than RAID 5.
- Important for avoiding **latent block + disk failure loss**.

2. Hardware vs. Software RAID

Software RAID

- Implemented entirely in software.
- No dedicated hardware required.

Hardware RAID

- Uses special controllers and often **NVRAM**.
- Protects against power loss by recording in-progress writes.
- If power fails mid-write (e.g., in RAID 1), inconsistent data may occur.
- Recovery may require a **full scan**.

3. Reliability Considerations

Common Risks

- **Latent Sector Failures:** Previously stored data becomes unreadable.
- **Silent Data Corruption:** Occurs without immediate detection.

Protective Measures

- **Data Scrubbing:** Continuous scans to detect and fix bad sectors.
- **Hot Swapping:** Replace disks while system is online.
- **Spare Disks:** Auto-replacement reduces rebuild time.
- **Redundancy:** Multiple power supplies and network controllers.

4. Performance Optimization

Disk Access Strategies

- **Buffering:** Cache blocks in memory.
- **Read-Ahead:** Pre-load blocks likely to be needed.
- **Disk-Arm Scheduling:** Minimize head movement (elevator algorithm).

File Organization

- Allocate blocks contiguously when possible.
- **Fragmentation** increases seek times.
- Defragmentation tools reorganize blocks for speed.

Non-Volatile Write Buffers

- Ensure consistency across mirrored/striped arrays.
- Protect against sudden power failure.